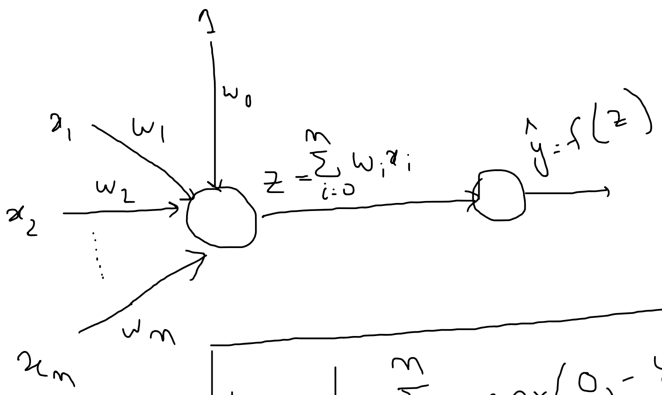
Perceptron → Divides the region



$L = f(w_0, w_1, \dots w_n)$ ← minimize

$n = \#$row

$$\boxed{L = \frac{1}{n}\sum_{i=1}^{n} \max\left(0, -y_i f(x_i)\right)}$$

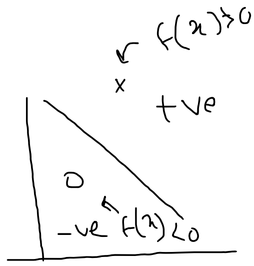
OLS ✗ →

GD →

$f(x_i) = w_0 x_{i0} + w_1 x_{i1} + w_2 x_{i2} + \dots + w_n x_{in}$

for i^{th} row

hinge loss



y	$\hat{y}$
+ve	+ve → 0
+ve	-ve → $y_i f(x_i)$
-ve	+ve → $y_i f(x_i)$
-ve	-ve → 0

$$W = W - \eta \frac{dL}{dW}$$

$$\boxed{W \rightarrow f(x_i) \rightarrow L}$$

$$\boxed{W \longrightarrow L} ✗$$

$w_0 = w_0 - \eta \frac{dL}{dw_0}$ $\quad w_1 = w_1 - \eta \frac{dL}{dw_1}$ $\quad \dots \quad w_n = w_n - \eta \frac{dL}{dw_n}$

$$\frac{dL}{dw_0} = \frac{dL}{d f(x_i)} \cdot \frac{d\, f(x_i)}{dw_0}$$

⇓

$$\frac{d}{d f(x_i)} \max(0, -y_i \underbrace{f(x_i)}_{z})$$

$= 0$ $\qquad = -y_i f(x_i)$

⇓

$$\frac{d\,[f(x_i)]}{dw_0} = \frac{d}{dw_0}\left(w_0 + \underbrace{w_1 x_{i1} + \dots + w_n x_{in}}\right)$$

$= 1$

$= 0$ $\qquad = -y_i f(x_i)$

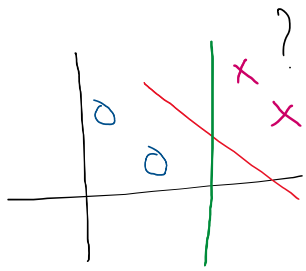
$$\frac{d\,0}{df(x_i)} \qquad \frac{d}{df(x_i)}\,-y_i f(x_i) \qquad \left|\begin{array}{l} =0 \\ =0 \end{array}\right| = -y_i$$

$$=0 \qquad = -y_i$$

$$\frac{dL}{dw_1} = \frac{dL}{df(x_i)} \cdot \frac{df(x_i)}{dw_1} \begin{cases} = 0 & 0 \\ = \cdot y_i f(x_i) = -y_i x_{i1} \end{cases}$$

$$\frac{dL}{dw_n} \begin{cases} = 0 & 0 \\ = \cdot y_i f(x_i) = -y_i x_{in} \end{cases}$$

$\boxed{\text{Perceptron trick}}$ ← random



$$=0$$

$$w_0 = w_0 - \eta \frac{dL}{dw}$$

$$\Rightarrow \boxed{w_0 = w_0}$$

$$w_0 = w_0 - \eta(-y_i)$$

$$\underset{w_1}{A}x + \underset{w_2}{B}y + \underset{w_0}{C} = 0 \quad \leftarrow \quad \Big| \; w_1x_1 + w_2x_2 + w_0 = 0$$

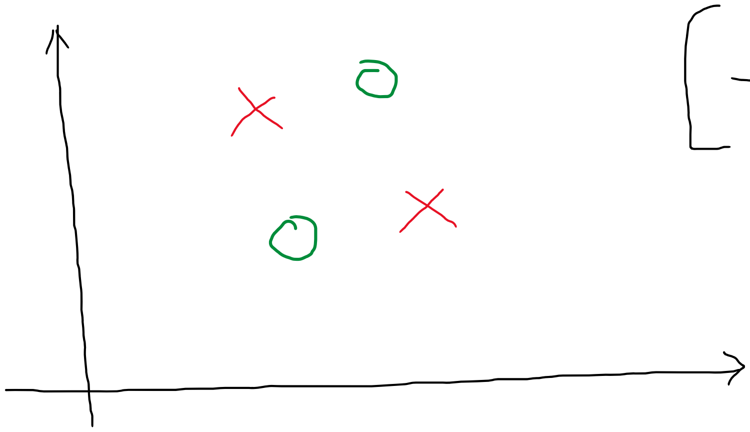
$$By = -Ax - C$$

$$y = -\frac{A}{B} - \frac{C}{B}$$

$$m = -\frac{w_1}{w_2} \qquad c = \frac{w_0}{w_2}$$

Problem

30 November 2025 18:31

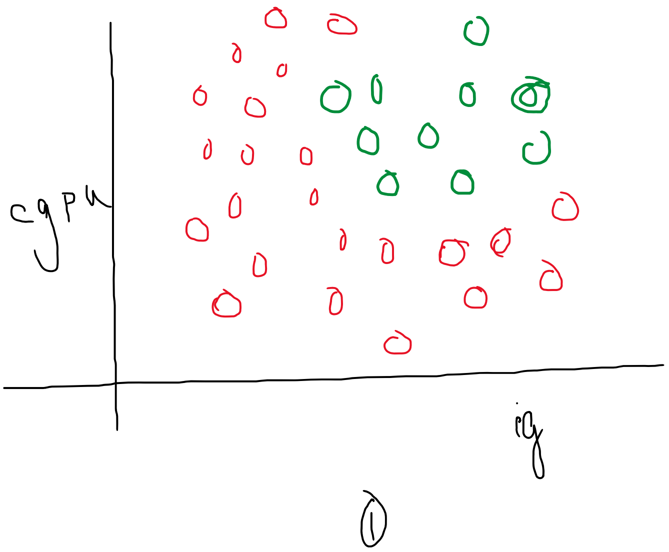


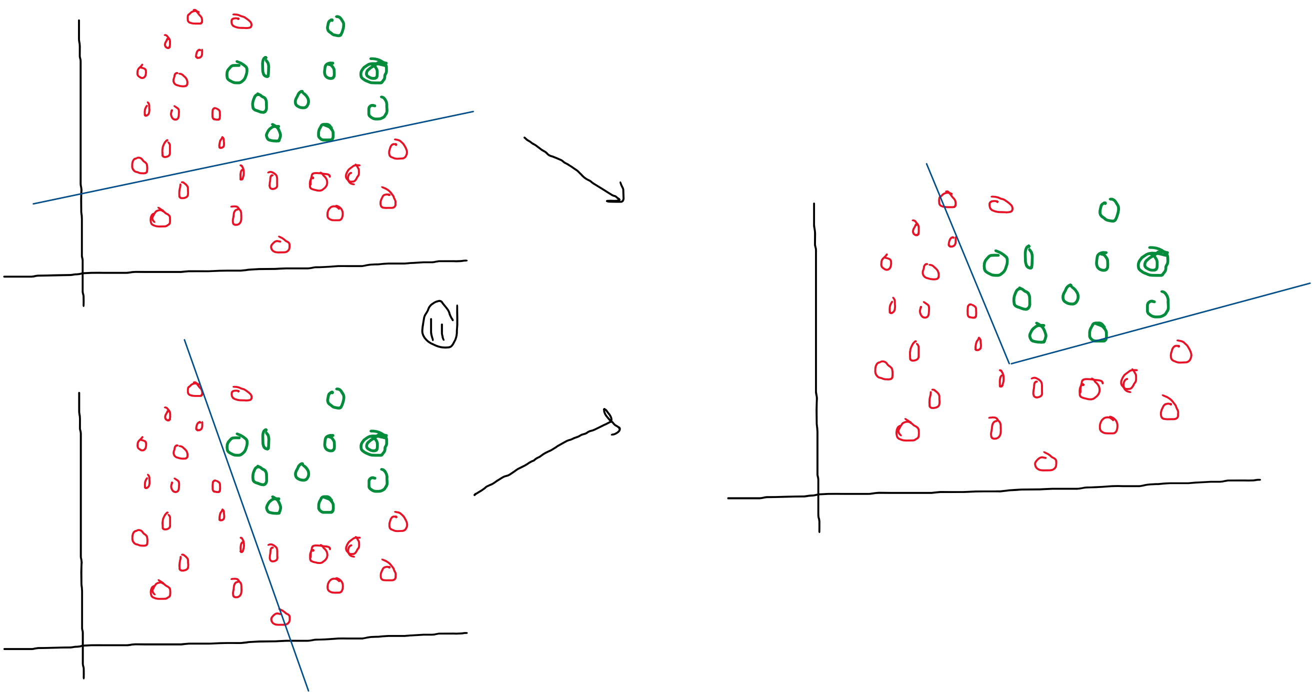
[Non Linear Dataset cannot be classified]

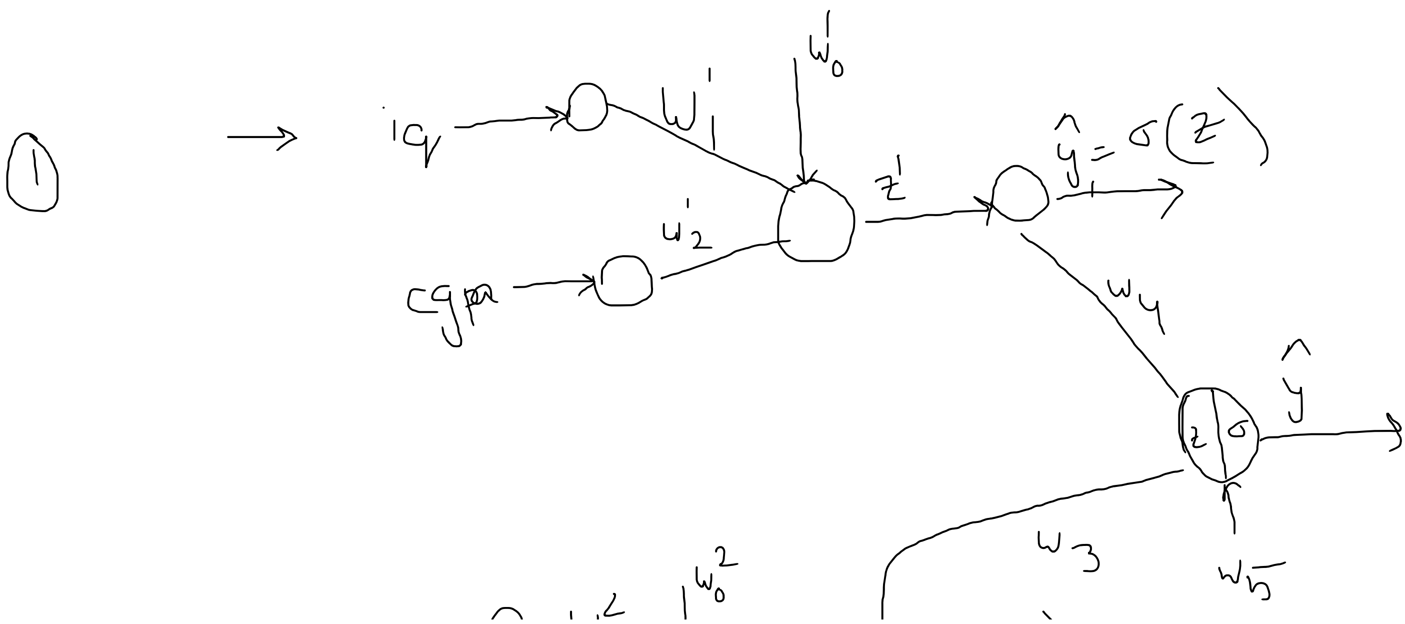
Solution → [Use Multiple perceptron]

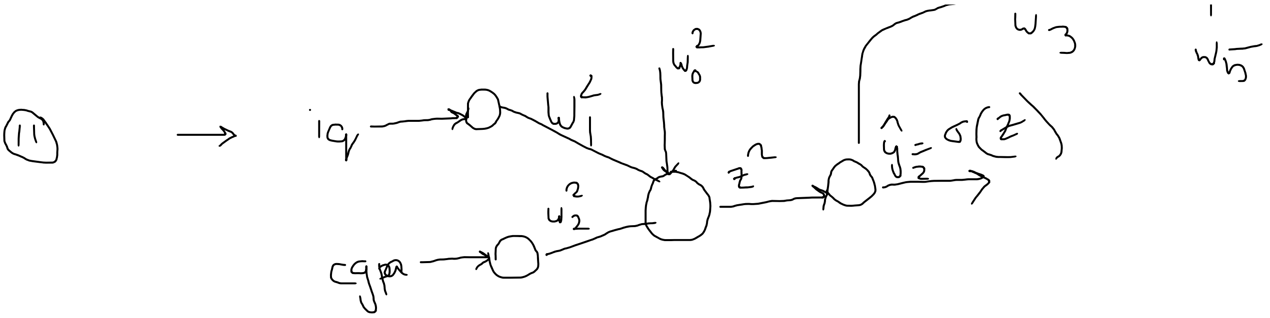
Multi Layer Perceptron

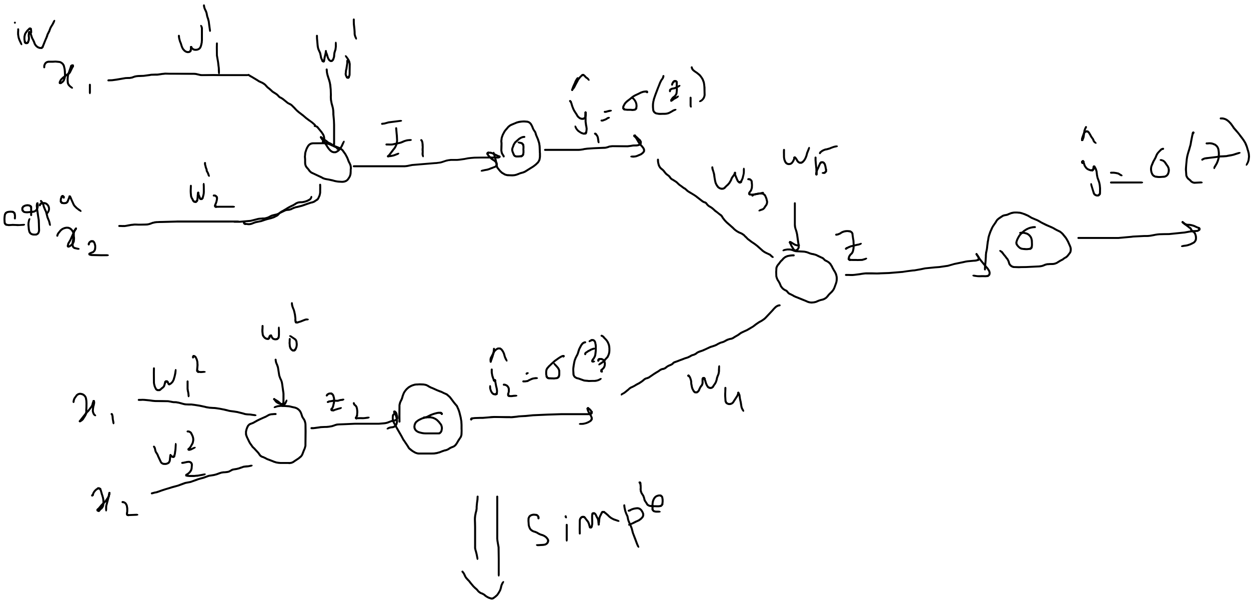
30 November 2025 18:34

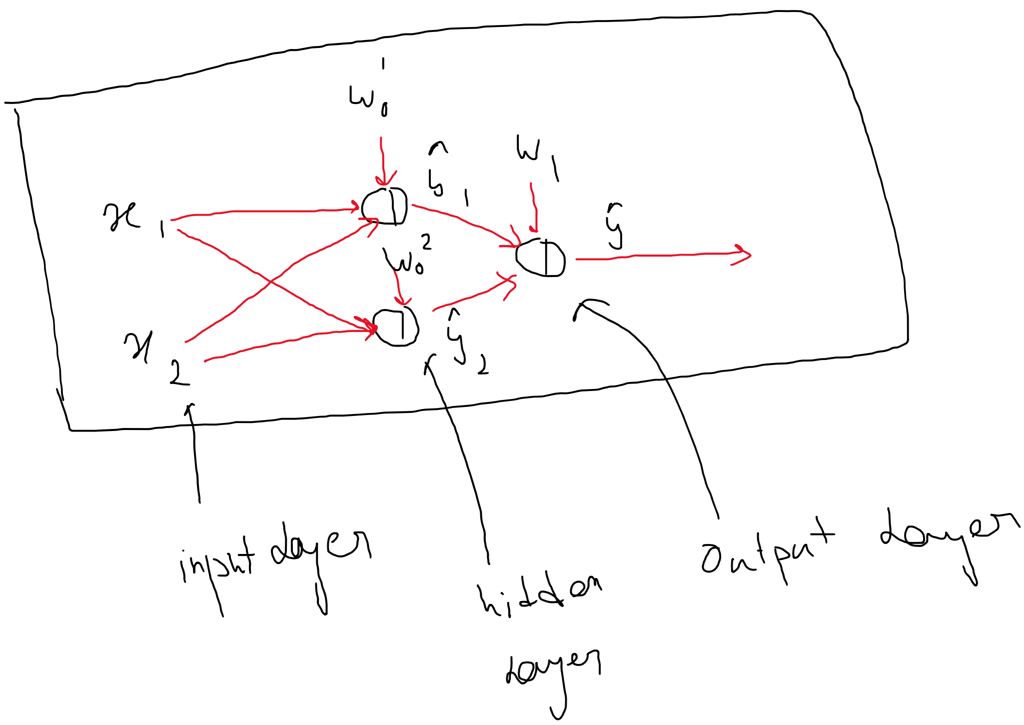












$\boxed{\sigma}$

$$\sigma = \frac{e^{-z}}{1+e^{-z}}$$

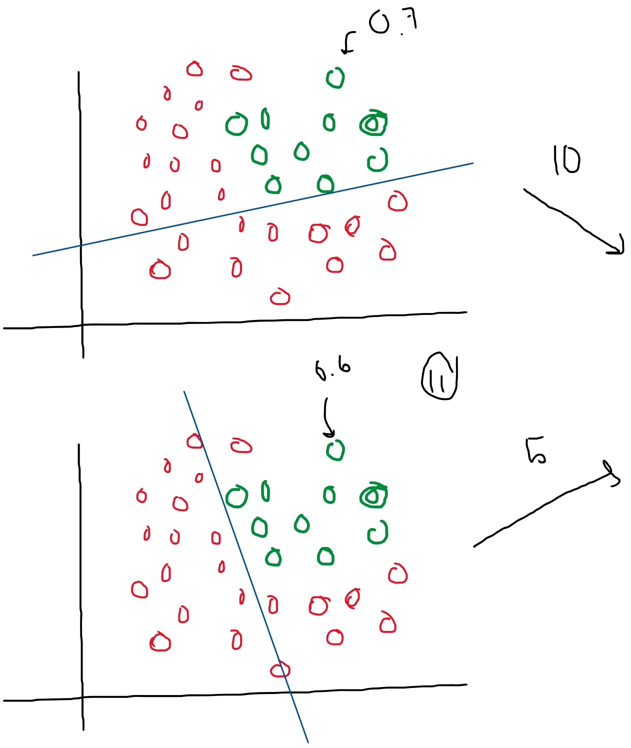
relu

$$= \max(0, z)$$

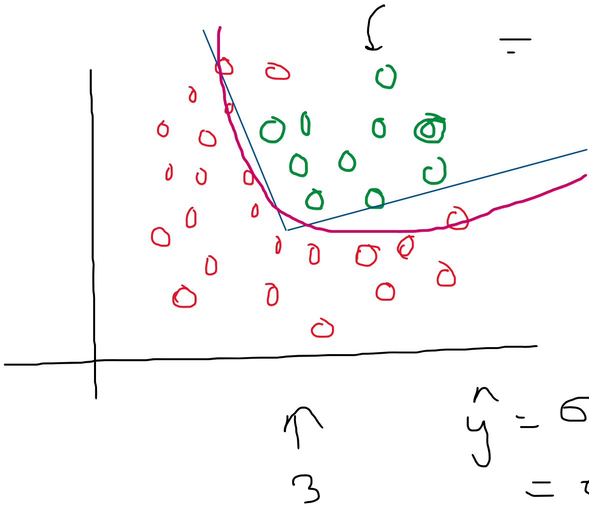
tanh

$$= \frac{1-x}{1+x}$$

https://playground.tensorflow.org/

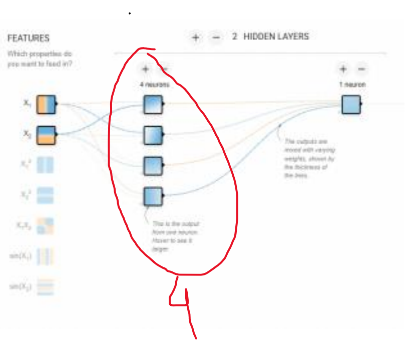


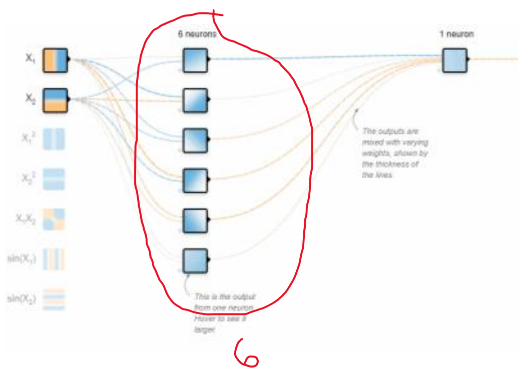
$z = 0.7 \times 10 + 0.6 \times 5 + 3 = 13$



$\hat{y} = \sigma(z) = \sigma(13) \simeq (0,1)$

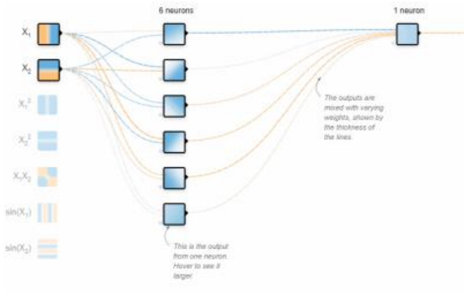
I) Increase nodes in hidden layer

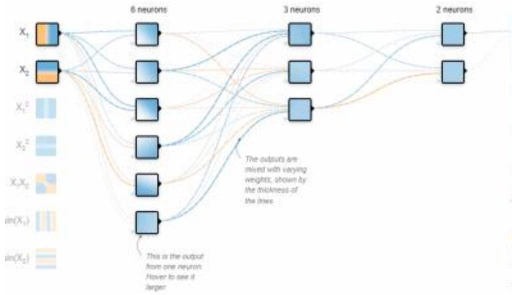




II) Adding more nodes in i/p column

III) Increase number of hidden layers.


6 neurons
1 neuron
X1
X2
The outputs are mixed with varying weights, shown by the thickness of the lines.
This is the output from one neuron. Hover to see it larger.

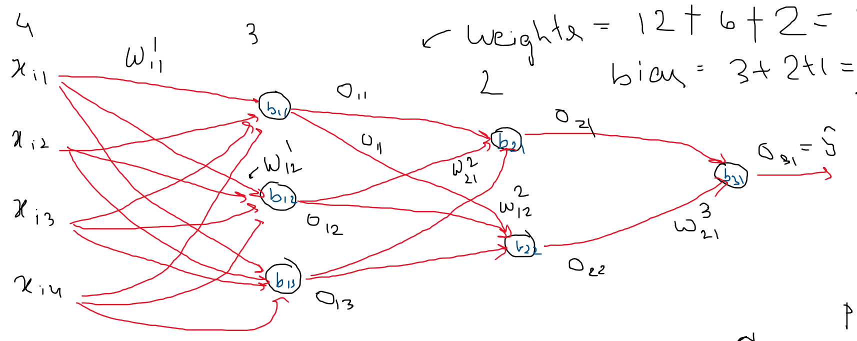

6 neurons
3 neurons
2 neurons
X1
X2
The outputs are mixed with varying weights, shown by the thickness of the lines.
This is the output from one neuron. Hover to see it larger.

Forward Propagation

30 November 2025 18:54

cgpa	iq	10^{th} Marks	12^{th} Marks	placed
7.2	72	492	315	0
9.64	148	608	436	1

Forward propagation is the process in a neural network where input data is passed through the network's layers to generate an output or prediction



← weights = 12 + 6 + 2 = 20

bias = 3 + 2 + 1 = 6

26

trainable parameters

Input layer (layer 0) | layer 1 | layer 2 — Hidden layers | layer 3 — o/p layer

W^k_{ij} — k ← o/p layer; j ← node number of o/p; i ↑ node number of i/p

layer #1 :-

$$O_{11} = b_{11} + w^1_{11} x_{i1} + w^1_{21} x_{i2} + w^1_{31} x_{i3} + w^1_{41} x_{i4}$$

$$\underbrace{\begin{bmatrix} w^1_{11} & w^1_{12} & w^1_{13} \\ w^1_{21} & w^1_{22} & w^1_{23} \\ w^1_{31} & w^1_{32} & w^1_{33} \\ w^1_{41} & w^1_{41} & w^1_{43} \end{bmatrix}}_{(4\times3)} \quad \underbrace{\begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ x_{i4} \end{bmatrix}}_{(1\times4)\ (4,1)}$$

$$\underbrace{\begin{bmatrix} w^1_{11} & w^1_{12} & w^1_{13} \\ w^1_{21} & w^1_{22} & w^1_{23} \\ w^1_{31} & w^1_{32} & w^1_{33} \\ w^1_{41} & w^1_{41} & w^1_{43} \end{bmatrix}^T}_{(3,4)} \times \underbrace{\begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ x_{i4} \end{bmatrix}}_{(4,1)} + \underbrace{\begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}}_{(3,1)} = \underbrace{\begin{bmatrix} \cancel{O_{11}}\ z_{11} \\ \cancel{O_{12}}\ z_{12} \\ \cancel{O_{13}}\ z_{13} \end{bmatrix}}_{(3,1)}$$

$\begin{bmatrix} O_{11} \\ O_{12} \\ O_{13} \end{bmatrix}$ ← σ / relu / linear ← activation function $\hat{y} = ?$

$\hat{y} = \sigma(z)$ $\quad \hat{y} = relu(z)$

layer 2

$$\begin{bmatrix} w^2_{11} & w^2_{12} \\ w^2_{..} & w^2_{..} \end{bmatrix} \quad \begin{bmatrix} O_{11} \\ O_{11} \end{bmatrix} + \begin{bmatrix} b_{21} \end{bmatrix}$$

$$\begin{bmatrix} w^2_{11} & w^2_{12} \\ w^2_{21} & w^2_{22} \\ w^2_{31} & w^2_{32} \end{bmatrix} \begin{bmatrix} O_{11} \\ O_{12} \\ O_{13} \end{bmatrix} + \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}$$

$(2,3)$ $(3,1)$ $(2,1)$

$$\begin{bmatrix} z_{21} \\ z_{22} \end{bmatrix} \xrightarrow{\sigma} = \begin{bmatrix} O_{21} \\ O_{22} \end{bmatrix}$$

Layer 3

$$\begin{bmatrix} w^3_{11} \\ w^3_{21} \end{bmatrix} \begin{bmatrix} O_{21} \\ O_{22} \end{bmatrix} + \begin{bmatrix} b_{31} \end{bmatrix} = \begin{bmatrix} z_{31} \end{bmatrix} \xrightarrow{\sigma} \begin{bmatrix} O_{31} \end{bmatrix}$$

$$a^{[1]} = \sigma\left(\underline{a^{[0]}} w^{[1]} + b^{[1]}\right)$$

Pred

$$\underbrace{\sigma\left(\underbrace{\sigma\left(\underbrace{\sigma\left(a^{[0]} w^{[1]} + b^{[1]}\right)}_{a^{[1]}} \times w^{[2]} + b^{[2]}\right)}_{a^{[2]}} \times w^{[3]} + b^{[3]}\right)}_{a^{[3]}}$$